



University of Gondar

College of Informatics

Department of Computer Science

Computer Vision and Image Processing (CoSc4113)

Chapter one: Introduction to Computer Vision and Image Processing

Objectives

By the end of this lesson, you will be able to:

- 1 Define what computer vision and image are, and explain the difference between them.
- 2 Identify related fields like AI and robotics, and describe how they connect to computer vision.
- 3 Give examples of real-world applications of computer vision and image processing.
- 4 List and explain the main steps in the image processing workflow

Contents

- 1 What is a computer vision?
- 2 What is image?
- 3 Related fields in CV
- 4 Computer Vision Vs image processing
- 5 Application of CV and IP
- 6 Different Image processing examples
- 7 Fundamental steps in image processing

1 What is a computer vision?

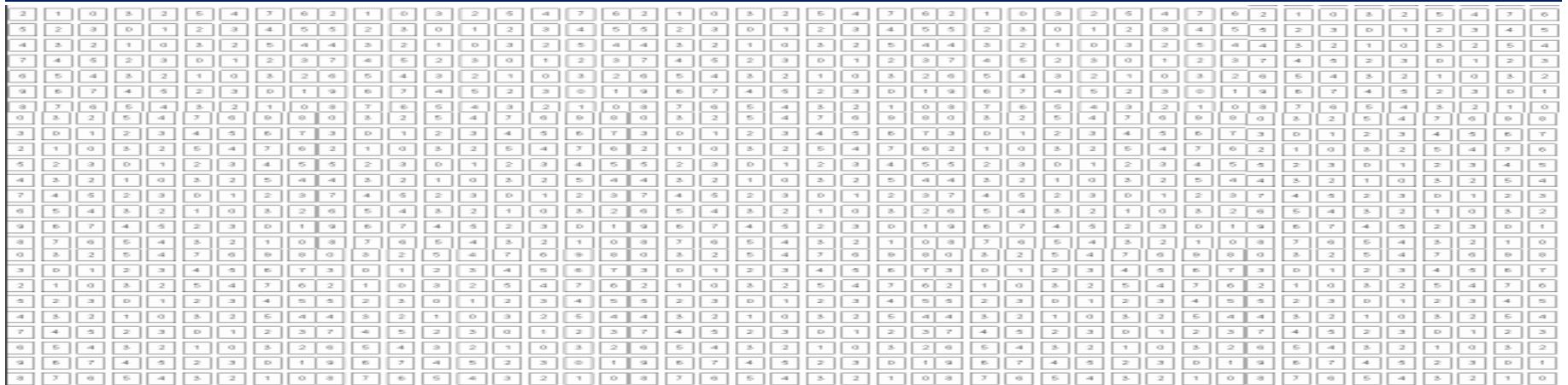
- ❖ It is a field of artificial intelligence that enables machines to see, analyze, and interpret the visual world like humans.
- ❖ It is the process of using artificial intelligence to enable computers to obtain meaningful data from visual inputs
- ❖ It can also be defined as a field of study that seeks to develop techniques to help computers/Machine “see” and understand the content of digital images such as photographs and videos
- ❖ By using images or videos, or 3D scans CV systems extract meaningful information to make decisions or provide outputs.
- ❖ The goal is to mimic human vision by extracting high-level understanding from pixels

1 What is a computer vision?

- ❖ From the perspective of engineering, it seeks to automate tasks that the human visual system can do
- ❖ The overall goal of computer vision is to give computers (super) human-level perception

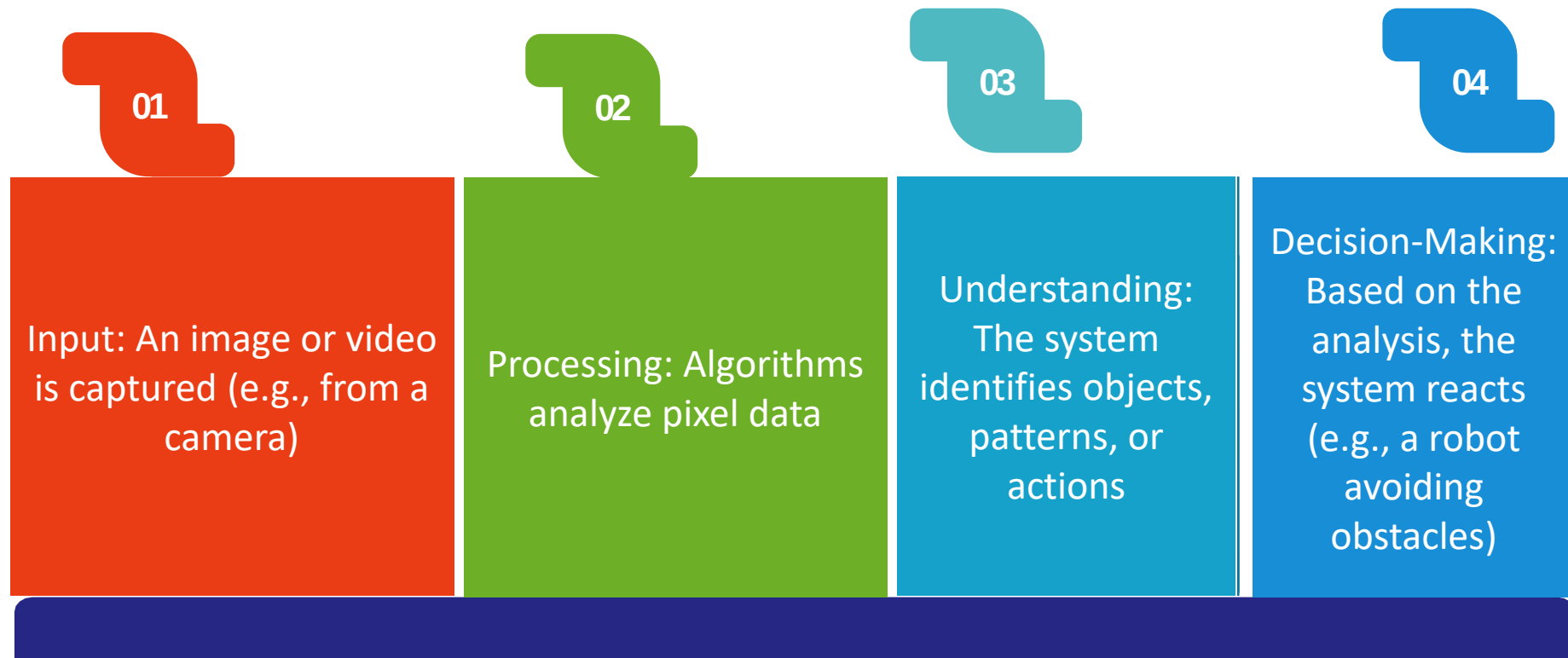


❖ Example



What a computer sees

1 How Does Computer Vision Work?



2 What is an Image?

- ❖ An **image** is a 2D array of **pixels**, where each pixel represents the intensity of light at that location.
- ❖ It is a two-dimensional function, where x and y are *spatial* (plane) coordinates, and the amplitude of f at any pair of coordinates (x, y) is called the *intensity* or *gray level* of the image at that point.
- ❖ When x , y , and the intensity values of f are all finite, discrete quantities, we call the image a digital image.
- ❖ Pixels or picture elements (image element) is the smallest unit in an image

❖ Representing Digital Images

- Digital image is represented by an $M \times N$ numerical array as

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & \dots & f(1,N-1) \\ \dots & \dots & \dots & \dots \\ f(M-1,0) & f(M-1,1) & \dots & f(M-1,N-1) \end{bmatrix}$$

3 Related Fields in Computer Vision

Image
Processing

Focuses on enhancement and transformation of images (e.g., filtering, contrast adjustment)

Pattern
Recognition

Identifies patterns in data (used in handwriting recognition, fingerprint matching)

Machine
Learning &
Deep Learning

Enables learning from data for tasks like classification or object detection. Example: A CNN learns to distinguish cats from dogs by analyzing thousands of labeled images.

Robotics

Uses vision for navigation, object manipulation, and decision making using cameras and sensors. Example: A warehouse robot avoiding obstacles while moving goods.

Artificial
Intelligence

Enables decision-making based on visual data. Example: A surveillance system detecting suspicious activity.

Augmented
Reality &
Virtual Reality

Overlays digital objects on real-world images. Example: Snapchat filters, virtual furniture placement in IKEA's app.

4 Computer Vision Vs. Image Processing

	Image Processing	Computer Vision
Goal	Enhance image quality	Understand and interpret image
Focus	processing the raw input images to enhance them or preparing them to do other tasks Low-level pixel operations	High-level semantic understanding
Example	Denoising, sharpening, Rescaling image (Digital Zoom), Correcting illumination, Changing tones	Object detection, scene labeling, Face detection, Hand writing recognition

5 Application of Computer Vision and Image Processing

Computer Vision

- ✓ Facial recognition (security, authentication)
- ✓ License plate recognition
- ✓ Autonomous vehicles
- ✓ Document image analysis
- ✓ Industrial inspection and robotics
- ✓ Remote sensing applications: Plotting weather maps, Oil exploration.
- ✓ Vision used for control: Road traffic monitoring, passive surveillance, etc.
- ✓ Medical Imaging

Image Processing

- ✓ Image enhancement for satellite images
- ✓ Restoration of old/damaged photos
- ✓ Noise filtering in medical images
- ✓ Preprocessing for OCR systems
- ✓ Photography- Enhancing low-light images. E.g. Night mode in smartphone cameras.
- ✓ Forensics- Enhancing blurry surveillance footage. License plate recognition

5 Application of Computer Vision and Image Processing

❖ In Agriculture

- ✓ Today, farmers are leveraging computer vision to enhance agricultural productivity.
- ✓ Companies specializing in agriculture technology are developing advanced computer vision and artificial intelligence models for *sowing and harvesting purposes*.
- ✓ These solutions are also useful for *weeding, detecting plant health, and advanced weather analysis*.
- ✓ Computer vision has numerous existing and upcoming applications in agriculture, including *drone-based crop monitoring, automatic spraying of pesticides, yield tracking, and smart crop sorting & classification*.
- ✓ for further analysis, these AI-powered solutions scan the *crops' shape, color, and texture*.
- ✓ Through computer vision technology, weather records, forestry data, and field security are also increasingly used

5 Application of Computer Vision and Image Processing

❖ In Autonomous vehicles

- ✓ Market leaders such as Tesla, backed by advanced technologies such as computer vision and 5G, are making great strides.
- ✓ Tesla's autonomous cars use multi-camera setups to analyze their surroundings.
- ✓ This enables the vehicles to provide users with advanced features, such as autopilot.
- ✓ The vehicle also uses 360-degree cameras to detect and classify objects through computer vision.
- ✓ Drivers of autonomous cars can either drive manually or allow the vehicle to make autonomous decisions.
- ✓ In case a user chooses to go with the latter arrangement, these vehicles use computer vision to engage in advanced processes such as path planning, driving scene perception, and behavior arbitration

5 Application of Computer Vision and Image Processing

❖ In Face Recognition

- ✓ While facial recognition is already in use at the personal level, such as through smartphone applications, the public security industry is also a noteworthy driver of facial detection solutions.
- ✓ ***How Facial recognition can work?***
- ✓ Detecting and recognizing faces in public is a contentious application of computer vision that is already being implemented in certain jurisdictions and banned in others.
- ✓ Successful facial detection relies on deep learning and machine vision. Proponents support computer vision-powered facial recognition because it can be useful for detecting and preventing criminal activities. These solutions also have applications in tracking specific persons for security missions

5 Application of Computer Vision and Image Processing

❖ In Interactive Entertainment

- ✓ Gone are the days when digital entertainment meant that the viewer had to sit and watch without participating.
- ✓ Today, interactive entertainment solutions leverage computer vision to deliver truly immersive experiences.
- ✓ Cutting-edge entertainment services use artificial intelligence to allow users to partake in dynamic experiences.
- ✓ For instance, Google Glass and other smart eyewear demonstrate how users can receive information about what they see while looking at it. The information is directly sent to the user's field of vision. These devices can also respond to head movements and changes in expressions, enabling users to transmit commands simply by moving their heads

5 Application of Computer Vision and Image Processing

❖ In Medical Imaging

- ✓ Medical systems rely heavily on pattern detection and image classification principles for diagnoses.
- ✓ While these activities were largely carried out manually by qualified healthcare professionals, computer vision solutions are slowly stepping up to help doctors diagnose medical conditions.
- ✓ There has been a noteworthy increase in the application of computer vision techniques for the processing of medical imagery.
- ✓ This is especially prevalent in pathology, radiology, and ophthalmology.
- ✓ Visual pattern recognition, through computer vision, enables advanced products, such as Microsoft InnerEye, to deliver swift and accurate diagnoses in an increasing number of medical specialties

5 Application of Computer Vision and Image Processing

❖ In Education

- ✓ With remote education receiving a leg-up due to the COVID-19 pandemic, the education technology industry is also leveraging computer vision for various applications.
- ✓ For instance, teachers use computer vision solutions to evaluate the learning process non-obstructively.
- ✓ These solutions allow teachers to identify disengaged students and tweak the teaching process to ensure that they are not left behind.
- ✓ Apart from this, AI vision is being used for applications such as school logistic support, knowledge acquisition, attendance monitoring, and regular assessments.
- ✓ One common example of this is computer vision-enabled webcams, which are being used to monitor students during examinations

5 Application of Computer Vision and Image Processing

❖ In Transportation

- ✓ Computer vision systems are being increasingly applied to increase transportation efficiency.
- ✓ For instance, computer vision is being used to detect traffic signal violators, thus allowing law enforcement agencies to minimize unsafe on-road behavior.
- ✓ Intelligent sensing and processing solutions are also being used to detect speeding and wrong-side driving violations, among other disruptive behaviors.
- ✓ Apart from this, computer vision is being used by intelligent transportation systems for *traffic flow analysis*

5 Application of Computer Vision and Image Processing

❖ Face Detection



Sony Cyber-shot



Age recognition



Smile recognition

❖ Complex Decision

First-down line



6 Different Image Processing Examples

- ❖ **Histogram Equalization** – Improves contrast for Poor contrast (too dark/bright) images by redistributing pixel intensities for better visibility.
- ❖ **Filtering** – Smoothen or sharpens images (e.g., Gaussian, Median filters)
- ❖ **Edge Detection** – Identifying object boundaries (algorithms like , Sobel, Prewitt, Canny)
- ❖ **Morphological Operations** – Cleaning up binary images (e.g., erosion, dilation)
- ❖ **Thresholding** – Converts grayscale to binary using a set value (e.g. Otsu's method** automatically finds the best threshold.)
- ❖ **Color Space Conversion** – RGB $\leftrightarrow$ HSV, YCbCr for segmentation or compression **Example:** Converting an RGB image to grayscale for edge detection
- ❖ Sharpening:- Used for filtering an images to enhance edges and ignore noise
- ❖ Smoothing:- remove noise. Used for smooth an images to denoise and ignore edges
- ❖ Enhancing:
- ❖ Restoring:
- ❖ Blurring/ deblurring

6 Fundamental Steps in Image Processing

1 Image Acquisition

Capturing an image using cameras, sensors or scanners. Digitization is necessary because the standard video signal is in analog (continuous) form and the computer requires a digitized or sampled version of that continuous signal

4 Feature Extraction

Extracting significant characteristics (edges, corners, textures)

2 Preprocessing

Enhancing image quality, removing noise, normalization, resizing

5 Classification /Object Recognition

Classifying or identifying objects in an image Using ML models

3 Segmentation

Dividing image into meaningful regions (e.g., foreground/background)

6 Interpretation & Decision-Making (Post-processing)

Improving the presentation, interpreting the results or acting on the analysis (e.g., a robot avoiding obstacles)

Thank You